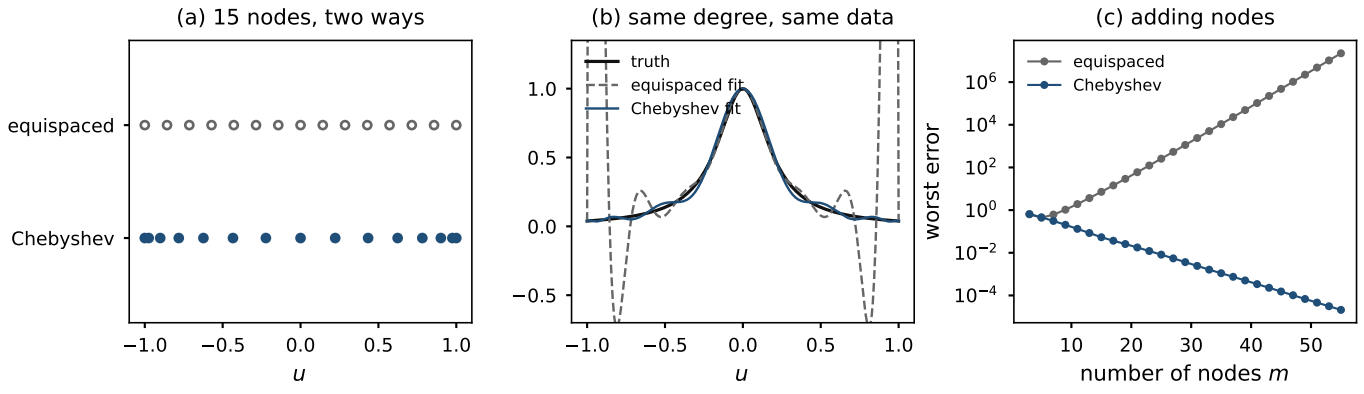
(a) 15 nodes, two ways
equispaced
Chebyshev
−1.0
−0.5
0.0
0.5
1.0
u
(b) same degree, same data
truth
equispaced fit
Chebyshev fit
1.0
0.5
0.0
−0.5
−1.0
−0.5
0.0
0.5
1.0
u
(c) adding nodes
equispaced
Chebyshev
worst error
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10
20
30
40
50
number of nodes m